Computational Maths

Sheet 1

1. Take $N=0$ $\rightarrow F_{N+2}=1$, $F_{N+1}=1$

In this case 0 divisions are required so b is already the GCD of a and b. Hence the only restriction is $a \geq 1 = F_{N+2}$, $b \geq 1 = F_{N+1}$ as required.

Assume that if N divisions are required for the pair a, b, then $a \geq F_{N+2}$ and $b \geq F_{N+1}$.

Take some a' and b' s.t. $N+1$ divisions are required.

We can write $a' = q b' + r$. The pair b' and r require $N+1$ divisions to find a GCD, so we can write $b' \geq F_{N+2}$ and $r_0 \geq F_{N+1}$

$\therefore a' \geq q F_{N+2} + F_{N+1} \geq F_{N+2} + F_{N+1}$ as $q \geq 1$.

$a' \geq F_{N+3}$ (and from before $b' \geq F_{N+2}$)

By induction the result is true.

2b. $p=137$ $q=173$ $e=7$.

$$\lambda(n) = \frac{(136)(172)}{\gcd(136,172)} = \frac{(136)(172)}{4} = 5848$$

We want $7d = 1 + 5848k$ $k \in \mathbb{Z}$

$7d + 5848k = 1$ Using the implementation with $a=7$, $b=5848$

$d = 1671$ $k = -2$